

HIGGINS DECOMPOSITION (H^S)

Operationalizing compositional data analysis — a runnable standard for researchers and the AI assistants they choose

“Compositional monitoring of energy-mix drift on the simplex” • CoDaWork 2026 • Coimbra, Portugal • June 1–5
Peter Higgins • Rogue Wave Audio / Binaural Test Lab • Markham, Ontario, Canada



11 validated domains	101 reference datasets	44 orders of magnitude	3 IEEE-floor physics confirmations	22+66 slides — talk + cinema scroll	~220 glossary v3.0 entries
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What this is. Aitchison gave the field its geometry in 1986; CoDaWork has trained four decades of methodologists. H^S packages current and developing CoDa methods into a **runnable operational standard** — seven phases, two human gates, deterministic hash-chained output, identical results whether the keystrokes come from a researcher or from an AI assistant. The math is standard; the operational frame may be new.

Why operationalize compositional analysis

Measurability. Turns theoretical compositional structure into reproducible per-step diagnostics: helmsman (handedness), Power Share, Activation Coefficient, navigation bearing. Quantifiable, comparable, audit-able.	Consistency. Fixed schema (CNT v3.1.0 / CNQ v2.0.0), fixed pipeline, byte-identical re-runs across machines, OS, BLAS, and years. Same input, same output, always.	Hypothesis testing. The same engine that produces a finding produces the falsifiability frame (MC-4 — four named defeat paths) that would overturn it. No publication-without-falsification.
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Numerical and operational advantages over ad-hoc CoDa pipelines

- **Helmert-ILR orthonormal coordinates** — no choice-of-balance arbitrariness; deterministic basis across teams.
- **atan2 for helmsman and navigation angles** — wrap-safe across $\pm\pi$; no precision loss or sign jumps at the cycle boundary.
- **Hash-chained provenance** — SHA-256 from raw CSV through CNT JSON, plates, projector, and manuscript figure. A reviewer in 2030 can prove nothing changed.
- **IEEE-floor cross-platform determinism** — same input, same output, every machine, every time. Verified on Backblaze, Planck CMB, neutrino oscillations.
- **Coherent Range Doctrine (CRD-1.0)** — multi-carrier comparisons on intersection of all members' ranges; asymmetric-drift artifacts eliminated.

The three-layer operational standard

CNT v3.1.0 measure	Closure → CLR → Helmert-ILR → per-step compositional metrics, helmsman, Power Share, Activation Coefficient, navigation, diagnostics, hashes. Current engine source v3.2.0 adds <code>navigation_2d</code> for ILR-Helmert PCA barycenter trajectory.
CNQ v2.0.0 name the algebra	Quaternion-view dashboards and higher-order structure diagnostics (CHSH joint coherence, twin-quaternion factoring at D=8 with Tsirelson-bound respect) where dimension policy supports them. Algebraic companion to CNT.
CCTT v1.0 operationalize	The runnable standard. Seven phases (diagnose → adapter <i>gate</i> → engine → outputs → render → self-verify <i>gate</i> → present + journal). Two human gates; everything else deterministic. The repo trains both researcher and AI assistant — same protocol, identical hash-verifiable output.

Five viewpoints in the talk

- **Composition** — share each carrier has.
- **Helmsman** — largest CLR displacement at a step.
- **Helmsman trajectory** — when steering changes.
- **Power Share** — how much squared CLR motion each carrier did.
- **Activation Coefficient** — Power Share ÷ starting share = "yeast factor."

CCTT 7-phase protocol

- 1 Diagnose
- 2 Adapter (*gate*)
- 3 Engine
- 4 Outputs
- 5 Render
- 6 Self-verify (*gate*)
- 7 Present + journal

Operational evidence — what the standard surfaces: a carrier can be small in share but large in structural work. **USA Solar 2012 → 2013:** 0.107% starting share, 81.7% of structural Power Share, **Activation Coefficient ≈ 760x**. The cross-country deceptive-drift signature fires in **5 of 9 countries** (AUS, CHN, GBR, IND, JPN) and does *not* fire in DEU (annual), FRA, USA, or WLD. The protocol discriminates; it does not over-fire. A regression on raw shares would not have surfaced either finding.

Standard onboarding — choose your entry point

1. **Conference attendee:** CODA-Association/CONFERENCE_ATTENDEES.md — slide-by-slide follow-along.
2. **Explore visually (zero install):** CODA-Association/CODaWork2026/data_outputs/codawork2026_projector.html.
3. **Run your own composition:** QUICKSTART.md + ai-refresh/CCTT_QUICKSTART.md — 7-phase runbook, manual or AI-assisted.
4. **Verify a published number:** manuscript + Supplementary Information + per-country JSON + hash chain.
5. **Vocabulary lookup:** HCI-CNT/handbook/GLOSSARY.md v3.0 (~220 entries: PCA, SVD, CLR/ILR, Helmert, CHSH, Tsirelson, Activation Coefficient, MC-1..MC-4).

Talk	“Compositional monitoring of energy-mix drift on the simplex,” CoDaWork 2026, Coimbra, June 1–5. Find Peter during sessions and Q&A — happy to walk the projector live.
Contact	Peter Higgins — PeterHiggins@RogueWaveAudio.com • Rogue Wave Audio / Binaural Test Lab, Markham, Ontario, Canada
Repository	github.com/PeterHiggins19/higgins-decomposition (QR top-right) • community folder: CODA-Association/ • conference folder: CODA-Association/CODaWork2026/
How to cite	Higgins, P. (2026). <i>Compositional monitoring of energy-mix drift on the simplex</i> . CoDaWork 2026, Coimbra. Repo: github.com/PeterHiggins19/higgins-decomposition (commit in <code>HS_FAST_REFRESH.json</code>).
How to adopt	Fork the repo, run the 7-phase CCTT on your composition, file a <code>JOURNAL.md</code> . AI assistant follows the same gates. See <code>ai-refresh/COMMUNITY_TEST_PACKET.json</code> for the structured adoption test.
License	Apache-2.0 (code) • CC BY 4.0 (docs and figures). Fully open source — fork, audit, extend, attribute.

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line. The AI follows the same protocol.